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SDG 3 – DATA MANIPULATION AND ANALYSIS

Study Patient Health Records to Identify Lifestyle Factors Affecting Diabetes Prevalence

1. Introduction

Diabetes is a major public-health challenge and is closely associated with lifestyle, demographic, and health-related factors. Patient health records can provide valuable information for identifying patterns that may be associated with diabetes prevalence. This study uses data manipulation and analysis techniques such as NumPy, Pandas, DataFrames, descriptive statistics, aggregation, grouping, filtering, sorting, file input/output, and plotting to examine lifestyle factors related to diabetes. The analysis is aligned with Sustainable Development Goal 3 (SDG 3), which aims to ensure healthy lives and promote well-being for all at all ages.

2. Objective

The main objective is to analyze patient health records and identify lifestyle factors that are associated with diabetes prevalence. The study focuses on variables such as age, body mass index (BMI), physical activity, smoking, alcohol consumption, diet, blood pressure, and family history. The analysis can help reveal relationships and trends that may support preventive healthcare strategies.

3. Dataset and Variables

A suitable patient-health dataset may contain records for individual patients, with each row representing one patient and columns representing health or lifestyle attributes. Important variables include Patient ID, Age, Gender, BMI, Physical Activity Level, Smoking Status, Alcohol Consumption, Diet Quality, Blood Pressure, Family History of Diabetes, and Diabetes Status. The diabetes status can be represented as 1 for diabetic and 0 for non-diabetic patients.

4. Data Loading and Preparation

The dataset can be imported from a CSV file using Pandas. Initial inspection should include checking the number of rows and columns, data types, missing values, duplicate records, and unusual values. Missing numerical values may be handled using appropriate statistical measures such as the median, while categorical missing values may be replaced using the mode when justified. Duplicate records should be identified and removed to improve data quality.

5. Data Manipulation Using Pandas and NumPy

Pandas DataFrames provide an efficient structure for filtering, grouping, sorting, and aggregating patient records. NumPy can support numerical calculations such as averages, standard deviations, and percentage calculations. For example, the mean BMI can be calculated for diabetic and non-diabetic groups. Filtering can identify patients with high BMI, low physical activity, or other characteristics of interest.

6. Statistical Analysis

Descriptive statistics can be used to summarize the dataset. Important measures include mean, median, minimum, maximum, standard deviation, and frequency. Diabetes prevalence can be calculated as the percentage of patients classified as diabetic. Group-wise statistics can then be used to compare diabetes prevalence across age groups, BMI categories, physical activity levels, smoking status, diet quality, and other lifestyle factors.

7. Lifestyle Factors and Diabetes Prevalence

BMI is an important factor to investigate because higher BMI may be associated with increased risk of type 2 diabetes. Physical activity can also be examined because insufficient activity may be associated with poorer metabolic health. Smoking and excessive alcohol consumption can be evaluated as behavioral factors. Diet quality is another important variable because dietary patterns can influence body weight and blood glucose regulation. Age and family history should also be considered because they can influence diabetes risk independently of lifestyle.

8. Grouping and Aggregation

Grouping patient records allows diabetes prevalence to be compared across categories. For example, patients can be grouped into low, normal, overweight, and obese BMI categories. Similarly, physical activity can be grouped into low, moderate, and high activity levels. The diabetes rate for each group can then be calculated. This makes it easier to identify groups with comparatively higher or lower prevalence.

9. Filtering and Sorting

Filtering can be used to examine specific patient groups, such as patients with BMI above a selected threshold or patients reporting low physical activity. Sorting can arrange groups according to diabetes prevalence, average BMI, or other health indicators. These operations help identify patterns and prioritize factors for further investigation.

10. Data Visualization

Charts can make relationships easier to understand. Bar charts can compare diabetes prevalence across physical activity levels, smoking categories, diet-quality groups, or BMI categories. Histograms can show the distribution of age or BMI. Box plots can compare BMI distributions between diabetic and non-diabetic patients. A heatmap can be used to display correlations among numerical health variables. Visualizations should include clear titles, axis labels, and legends where appropriate.

11. Example Python Workflow

A typical workflow begins by importing Pandas, NumPy, and Matplotlib. The CSV file is loaded into a DataFrame, followed by inspection of the dataset and treatment of missing values. New categorical variables, such as BMI categories, can be created. GroupBy operations can then calculate diabetes prevalence for each lifestyle category. Finally, charts can be generated to communicate the results. The exact code should be adapted to the column names and structure of the available dataset.

12. Interpretation of Findings

If the analysis shows higher diabetes prevalence among patients with high BMI and low physical activity, these variables may be important indicators of lifestyle-related risk. If poorer diet quality is associated with higher prevalence, nutritional interventions may deserve attention. However, observed associations should not automatically be interpreted as causal relationships. Factors such as age, genetics, socioeconomic conditions, access to healthcare, and existing medical conditions can affect the results.

13. Real-World Applications

The analysis can support public-health planning, preventive screening, patient education, and personalized lifestyle interventions. Healthcare organizations can use aggregated insights to identify populations that may benefit from nutrition counseling, physical-activity programs, or early screening. Such analysis can contribute to evidence-based strategies for reducing the burden of diabetes and improving population health.

14. SDG 3 Relevance

This study directly supports SDG 3: Good Health and Well-Being. Identifying modifiable lifestyle factors associated with diabetes can contribute to disease prevention and healthier communities. Data-driven analysis can help healthcare systems monitor risk patterns, evaluate interventions, and allocate resources more effectively.

15. Conclusion

Patient health-record analysis provides a practical way to investigate factors associated with diabetes prevalence. Using NumPy, Pandas, DataFrames, statistics, aggregation, grouping, filtering, sorting, file I/O, and plotting, meaningful patterns can be extracted from healthcare data. BMI, physical activity, diet, smoking, alcohol consumption, age, and family history are useful variables for analysis. The results should be interpreted carefully, with attention to data quality and the difference between association and causation. Overall, this type of analysis demonstrates how data science can support SDG 3 by contributing to diabetes prevention and improved health decision-making.